

OSPF CONFIGURATION

Step 1 : Assign ip address in all cisco routers

R1

```
R1#
R1#sh ip int br
Interface                IP-Address      OK? Method Status Prot
ocol
Ethernet0/0              10.0.0.1        YES TFTP  up    up
Ethernet0/1              20.0.0.1        YES TFTP  up    up
Ethernet0/2              unassigned      YES TFTP  down  down
Ethernet0/3              unassigned      YES TFTP  down  down
R1#
```

R2

```
R2#
R2#sh ip int br
Interface                IP-Address      OK? Method Status Prot
ocol
Ethernet0/0              20.0.0.2        YES TFTP  up    up
Ethernet0/1              30.0.0.1        YES TFTP  up    up
Ethernet0/2              unassigned      YES TFTP  down  down
Ethernet0/3              unassigned      YES TFTP  down  down
R2#
```

R3

```
sh
R3#
R3#sh ip int br
Interface                IP-Address      OK? Method Status Prot
ocol
Ethernet0/0              40.0.0.1        YES TFTP  up    up
Ethernet0/1              30.0.0.2        YES TFTP  up    up
Ethernet0/2              unassigned      YES TFTP  down  down
Ethernet0/3              unassigned      YES TFTP  down  down
R3#
```

Step 2 : Check interface bandwidth

```
R1
R1(config)#
R1(config)#do sh int eth0/0
Ethernet0/0 is up, line protocol is up
  Hardware is AmdP2, address is aabb.cc00.0100 (bia aabb.cc00.0100)
  Internet address is 10.0.0.1/8
  MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Auto-duplex, Auto Speed, media type is unknown media type
  input flow-control is off, output flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:01:34, output 00:00:00, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
```

Step 3 : Enable ospf protocol in all cisco router

```
R1
R1(config)#
R1(config)#router ospf 1
R1(config-router)#
R1(config-router)#router-id 1.1.1.1
R1(config-router)#
R1(config-router)#network 10.0.0.0 0.255.255.255 area 1
R1(config-router)#network 20.0.0.0 0.255.255.255 area 1
R1(config-router)#exit
R1(config)#
*Sep 23 03:56:22.031: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config)#

R2
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#
R2(config)#router ospf 2
R2(config-router)#
R2(config-router)#router-id 2.2.2.2
R2(config-router)#
R2(config-router)#network 20.0.0.0 0.255.255.255 area 1
R2(config-router)#network 30.0.0.0 0.255.255.255 area 1
*Sep 23 03:56:35.205: %OSPF-5-ADJCHG: Process 2, Nbr 1.1.1.1 on Ethernet0/0 from
LOADING to FULL, Loading Done
R2(config-router)#network 30.0.0.0 0.255.255.255 area 1
R2(config-router)#exit

R3
R3(config)#
R3(config)#router ospf 3
R3(config-router)#
R3(config-router)#router-id 3.3.3.3
R3(config-router)#
R3(config-router)#network 30.0.0.0 0.255.255.255 area 1
R3(config-router)#network 30.0.0.0 0.255.255.255 area 1
*Sep 23 03:57:34.705: %OSPF-5-ADJCHG: Process 3, Nbr 2.2.2.2 on Ethernet0/1 from
LOADING to FULL, Loading Done
R3(config-router)#network 40.0.0.0 0.255.255.255 area 1
R3(config-router)#exit
R3(config)#
```

Step 4 :Check ospf neighbor table

```
R1#
R1#
R1#
R1#
R1#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
2.2.2.2          1     FULL/BDR        00:00:35    20.0.0.2     Ethernet0/1
R1#
```

```
R2
R2(config)#do sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
3.3.3.3          1     FULL/BDR        00:00:35    30.0.0.2     Ethernet0/1
1.1.1.1          1     FULL/DR         00:00:34    20.0.0.1     Ethernet0/0
R2(config)#
```

```
R3
R3(config)#do sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
2.2.2.2          1     FULL/DR         00:00:30    30.0.0.1     Ethernet0/1
R3(config)#
```

Step 5 : Check ospf link state database table

```
R1#sh ip ospf database

OSPF Router with ID (1.1.1.1) (Process ID 1)

Router Link States (Area 1)

Link ID      ADV Router   Age         Seq#         Checksum Link count
1.1.1.1      1.1.1.1     582        0x80000005  0x0017B8  2
2.2.2.2      2.2.2.2     523        0x80000006  0x003959  2
3.3.3.3      3.3.3.3     477        0x80000004  0x008805  2

Net Link States (Area 1)

Link ID      ADV Router   Age         Seq#         Checksum
20.0.0.1     1.1.1.1     582        0x80000001  0x00F81C
30.0.0.1     2.2.2.2     523        0x80000001  0x00AC52
R1#
```

```
R2#sh ip ospf database

OSPF Router with ID (2.2.2.2) (Process ID 2)

Router Link States (Area 1)

Link ID      ADV Router   Age         Seq#         Checksum Link count
1.1.1.1      1.1.1.1     602        0x80000005  0x0017B8  2
2.2.2.2      2.2.2.2     541        0x80000006  0x003959  2
3.3.3.3      3.3.3.3     495        0x80000004  0x008805  2

Net Link States (Area 1)

Link ID      ADV Router   Age         Seq#         Checksum
20.0.0.1     1.1.1.1     602        0x80000001  0x00F81C
30.0.0.1     2.2.2.2     541        0x80000001  0x00AC52
R2#
```

```
R3#sh ip ospf database

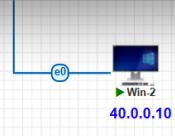
OSPF Router with ID (3.3.3.3) (Process ID 3)

Router Link States (Area 1)

Link ID      ADV Router   Age         Seq#         Checksum Link count
1.1.1.1      1.1.1.1     647        0x80000005  0x0017B8  2
2.2.2.2      2.2.2.2     587        0x80000006  0x003959  2
3.3.3.3      3.3.3.3     539        0x80000004  0x008805  2

Net Link States (Area 1)

Link ID      ADV Router   Age         Seq#         Checksum
20.0.0.1     1.1.1.1     647        0x80000001  0x00F81C
30.0.0.1     2.2.2.2     587        0x80000001  0x00AC52
R3#
```



Step 6 : Check interface cost of ospf

```
R1#sh ip ospf int eth0/0
Ethernet0/0 is up, line protocol is up
  Internet Address 10.0.0.1/8, Area 1, Attached via Network Statement
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 10
  Topology-MTID      Cost      Disabled      Shutdown      Topology Name
    0                10        no           no           Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 10.0.0.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
  Hello due in 00:00:01
```

```
R2#sh ip ospf int eth0/0
Ethernet0/0 is up, line protocol is up
  Internet Address 20.0.0.2/8, Area 1, Attached via Network Statement
  Process ID 2, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 10
  Topology-MTID      Cost      Disabled      Shutdown      Topology Name
    0                10        no           no           Base
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 20.0.0.1
  Backup Designated router (ID) 2.2.2.2, Interface address 20.0.0.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
  Hello due in 00:00:03
```

```
R3#sh ip ospf int eth0/0
Ethernet0/0 is up, line protocol is up
  Internet Address 40.0.0.1/8, Area 1, Attached via Network Statement
  Process ID 3, Router ID 3.3.3.3, Network Type BROADCAST, Cost: 10
  Topology-MTID      Cost      Disabled      Shutdown      Topology Name
    0                10        no           no           Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 3.3.3.3, Interface address 40.0.0.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
  Hello due in 00:00:08
```

Step 7 : Check routing table

